Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 4 · SLO 1 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 4 : SLO â€“ 1 | Stack Applications â€” Reversing a List and Balancing Symbols

I. Reversing a List Using a Stack

• Step 1: List A = [10, 20, 30, 40, 50] (5 elements).
• Step 2: Traverse A from the first element and push each onto Stack S:
• Push 10 -> S = [10]; Push 20 -> [10,20]; Push 30 -> [10,20,30];
• Push 40 -> [10,20,30,40]; Push 50 -> [10,20,30,40,50]. TOP = 50.
• Step 3: Pop from TOP one by one into array B:
• Pop 50 -> B[0]=50; Pop 40 -> B[1]=40; Pop 30 -> B[2]=30;
• Pop 20 -> B[3]=20; Pop 10 -> B[4]=10.
• Result: B = [50, 40, 30, 20, 10] â€” the reversed list.

II. Balanced Parenthesis Check: { P + (Q - R) }

• flag = 1. Scan left to right:
• '{' -> push. Stack: {
• '(' -> push. Stack: { (
• ')' -> pop '(' -> matches '('. Stack: {
• '}' -> pop '{' -> matches '{'. Stack: EMPTY.
• End of expression: stack empty and flag = 1 -> VALID (balanced).

III. Balanced Parenthesis Check: { D * (S + A))

• flag = 1. Scan left to right:
• '{' -> push. Stack: {
• '(' -> push. Stack: { (
• ')' -> pop '(' -> matches. Stack: {
• ')' -> pop '{' -> does NOT match ')' -> flag = 0.
• End of expression: flag = 0 -> INVALID (unbalanced).